Sekwang Lee

sekwang@postech.ac.kr
https://sekwang.vercel.app (Website)
https://www.linkedin.com/in/sekwang-lee/ (LinkedIn)

EDUCATION

Pohang University of Science and Technology (POSTECH) — From Mar 2027 (Expected)
M.S./Ph.D. Student in Computer Science and Engineering (CSE) — Pohang, Korea
- Advisor: Prof. Ilwoo Lyu

Pohang University of Science and Technology (POSTECH) — Feb 2021 – Feb 2027
B.S. in Computer Science and Engineering (CSE) — Pohang, Korea
- GPA: 3.92/4.3 **(Summa Cum Laude)**

KTH Royal Institute of Technology — Jan 2026 – Jun 2026
Exchange Student in Electrical Engineering and Computer Science (EECS) — Stockholm, Sweden

PUBLICATIONS

[C1] **"AdaPhase: Navigating the Trajectory with Adaptive Frequency Modulation for Content-preserving Stylization"**
IEEE/CVF Winter Conference on Applications of Computer Vision **(WACV)**, 2027
- **Sekwang Lee** and Ilwoo Lyu

RESEARCH EXPERIENCE

Undergraduate Research Intern, Medical Information Processing Lab - CG/CV Team — Jan 2025 – Feb 2027
Department of Computer Science and Engineering, POSTECH — Pohang, Korea
- Advisor: Prof. Ilwoo Lyu
- Reviewed literature on **spherical deep learning & cortical surface analysis**
- Conducted two research project: **"AdaPhase"** [C1] • **optimization-based spherical registration**

Undergraduate Research Intern, Computer Vision Lab — Aug 2023 – Nov 2023
Department of Computer Science and Engineering, POSTECH — Pohang, Korea
- Advisor: Prof. Suha Kwak
- Reviewed literature on **Fourier-based domain generalization** • Formulated research topic: **AdaIN-based Generalization**

WORK EXPERIENCE

Student Advisor — Feb 2025 – Dec 2025
Mueunjae School of Undergraduate Studies, POSTECH — Pohang, Korea
- Conducted **66 counseling sessions** for freshmen • TA for freshman courses: *Future Planning for College Life I & II*

Public Relations (PR) Coordinator — Sep 2024 – Dec 2025
Institute of Communication and Public Discourse, POSTECH — Pohang, Korea
- Led promotional content creation • Designed program posters **(founding member, invited by professor)**

Software Engineering Intern — Jun 2025 – Aug 2025
AI & Big Data Team, LG Electronics — Seoul, Korea
- Designed and implemented **"OneClick"**: a website task automation service **(founding member)**

Tech-Review Manager — Jul 2024 – Jul 2025
Start-up Support Team, POSTECH — Pohang, Korea
- Hosted technology seminars with startup CEOs • Organized a technology-based ideathon

Military Police (Military Service) — Jan 2022 – Jul 2023
Republic of Korea Army • Ministry of National Defense — Seoul, Korea
- Worked as squad leader (5 months)

PROJECTS

"Distillation Through Attention of SwinDETR" — Jul 2024 – May 2025

Undergraduate Research Program • *Deep Learning* (Graduate Course), POSTECH

- Worked as team leader (A+ in the *Deep Learning* course)
- Designed **SwinDETR** for small-object detection • Introduced **distillation queries**

"Autonomous Drone System with MIMO Signal Processing and Multimodal AI" — Mar 2024 – Dec 2024

I-Corps Program, Ministry of Science and ICT

- Worked as AI Developer
- Designed **FIFO-Transfuser (image-radar multimodal)** for foggy conditions • Conducted field research at two fire stations

AWARDS & HONORS

Best Research Project Award (Grand Prize) — Dec 2025

Departent of Computer Science and Engineering, POSTECH

- Conducted research project: **AdaPhase** [C1]

Next-generation Engineer Award — Aug 2025

Institute for Promotion of Engineering and Science of Korea (IPESK)

Presidential Science Scholarship — Jun 2025

Ministry of Science and ICT

Global Leadership Program — Sep 2024

Department of Computer Science and Engineering, POSTECH

- Merit-based scholarship for the top 8 CSE students

Book Review Essay Contest (Grand Prize) — Aug 2024

Institute of Communication and Public Discourse, POSTECH

- Wrote on the social embeddedness of science and scientists' responsibility for scientific uncertainty (writing (KOR))

IT & Game Scholarship — Jul 2024

Sangah Foundation

POSTECH Programming Contest (3rd Prize) — May 2024

Department of Computer Science and Engineering, POSTECH

Young Engineers Honor Society (YEHS) — Jan 2024

National Academy of Engineering of Korea (NAEK)

SPECIAL EXPERIENCE

Apple Foundation Program — Jun 2026 – Jul 2026

Apple Developer Academy @ POSTECH — Pohang, Korea

ARPA-E Energy Innovation Summit — Mar 2025

Advanced Research Projects Agency–Energy (ARPA-E) — Washington, D.C., USA

- Attended as a POSTECH undergraduate student

Korea Cross Country Cycling Road — Jul 2024

Ministry of Environment • Ministry of the Interior and Safety — Korea

- Cycled 633 km over 6 days from Busan to Incheon (completed the cross country cycling certification)

Self-Published Book: "I Can Live a Wonderful Day Today" (『나는 오늘도 멋진 날을 살 수 있다』) — Jun 2023 – Feb 2024

Self Project

- Authored a 190-page collection of stories and reflections during military service

CES 2024 — Jan 2024

Consumer Technology Association (CTA) — Las Vegas, USA

- Attended as a POSTECH undergraduate student